

For the use of a Registered Medical Practitioner Only

PRESCRIBING INFORMATION

ENHANCIN SUSPENSION

(Amoxicillin and Clavulanate Potassium for Oral Suspension USP)

COMPOSITION

ENHANCIN SUSPENSION

Each 5 ml of the constituted suspension contains:

Amoxicillin USP (as trihydrate)

Equivalent to amoxicillin.....250 mg

Potassium clavulanate

Equivalent to clavulanic acid62.5 mg

Excipients:

Butylated Hydroxytoluene, Xanthan Gum, Monosodium Citrate, Sodium Citrate, Passion Fruit Flavour, Aspartame, Sorbitol, Sodium Benzoate

DESCRIPTION

White to off - white powder forming a white to pale yellow suspension on constitution with water. The resulting suspension has a characteristic flavour.

PHARMACODYNAMIC AND PHARMACOKINETIC PROPERTIES

Pharmacodynamic properties

Pharmacotherapeutic group: Combinations of penicillins, incl. Beta-lactamase inhibitors; ATC code: J01CR02.

Mechanism of action

Amoxicillin is a semisynthetic penicillin (beta-lactam antibiotic) that inhibits one or more enzymes (often referred to as penicillin-binding proteins, PBPs) in the biosynthetic pathway of bacterial peptidoglycan, which is an integral structural component of the bacterial cell wall. Inhibition of peptidoglycan synthesis leads to weakening of the cell wall, which is usually followed by cell lysis and death.

Amoxicillin is susceptible to degradation by beta-lactamases produced by resistant bacteria and therefore the spectrum of activity of amoxicillin alone does not include organisms which produce these enzymes.

Clavulanic acid is a beta-lactam structurally related to penicillins. It inactivates some beta-lactamase enzymes thereby preventing inactivation of amoxicillin. Clavulanic acid alone does not exert a clinically useful antibacterial effect.

Pharmacokinetic/pharmacodynamic relationship

The time above the minimum inhibitory concentration (T>MIC) is considered to be the major determinant of efficacy for amoxicillin.

Mechanisms of resistance

The two main mechanisms of resistance to amoxicillin/clavulanic acid are:

- Inactivation by those bacterial beta-lactamases that are not themselves inhibited by clavulanic acid, including class B, C and D.
- Alteration of PBPs, which reduce the affinity of the antibacterial agent for the target.

Impermeability of bacteria or efflux pump mechanisms may cause or contribute to bacterial resistance, particularly in Gram-negative bacteria.

The prevalence of resistance may vary geographically and with time for selected species, and local information on resistance is desirable, particularly when treating severe infections. As necessary, expert advice should be sought when the local prevalence of resistance is such that the utility of the agent in at least some types of infections is questionable.

Commonly susceptible species

Aerobic Gram-positive micro-organisms

Enterococcus faecalis

Gardnerella vaginalis

Staphylococcus aureus (methicillin-susceptible)[£]

Coagulase-negative staphylococci (methicillin-susceptible)

Streptococcus agalactiae

*Streptococcus pneumoniae*¹

Streptococcus pyogenes and other beta-haemolytic streptococci

Streptococcus viridans group

Aerobic Gram-negative micro-organisms

Capnocytophaga spp.

Eikenella corrodens

*Haemophilus influenzae*²

Moraxella catarrhalis

Pasteurella multocida

Anaerobic micro-organisms

Bacteroides fragilis

Fusobacterium nucleatum

Prevotella spp.

Species for which acquired resistance may be a problem

Aerobic Gram-positive micro-organisms

Enterococcus faecium [§]

Aerobic Gram-negative micro-organisms

Escherichia coli

Klebsiella oxytoca

Klebsiella pneumoniae

Proteus mirabilis

Proteus vulgaris

Inherently resistant organisms

Aerobic Gram-negative micro-organisms

Acinetobacter sp.

Citrobacter freundii

Enterobacter sp.

Legionella pneumophila

Morganella morganii

Providencia spp.

Pseudomonas sp.

Serratia sp.

Stenotrophomonas maltophilia

Other micro-organisms

Chlamydophila pneumoniae

Chlamydophila psittaci

Coxiella burnetii

Mycoplasma pneumoniae

§ Natural intermediate susceptibility in the absence of acquired mechanism of resistance.

£ All methicillin-resistant staphylococci are resistant to amoxicillin/clavulanic acid

¹ *Streptococcus pneumoniae* that are resistant to penicillin should not be treated with this presentation of amoxicillin/clavulanic acid (see **DOSE AND METHOD OF ADMINISTRATION, WARNINGS AND PRECAUTIONS**).

² Strains with decreased susceptibility have been reported in some countries in the Europe with a frequency higher than 10%.

Pharmacokinetics

Absorption

Amoxicillin and clavulanic acid, are fully dissociated in aqueous solution at physiological pH. Both components are rapidly and well absorbed by the oral route of administration. Following oral administration, amoxicillin and clavulanic acid are approximately 70% bioavailable. The plasma profiles of both components are similar and the time to peak plasma concentration (T_{max}) in each case is reported to be approximately one hour.

The reported pharmacokinetic parameters of amoxicillin/clavulanic acid (500 mg/125 mg tablets three times daily) in fasting state are presented below:

Mean ($\pm$ SD) pharmacokinetic parameters

Active substance(s) administered	Dose	C_{max}	T_{max}^*	$AUC_{(0-24h)}$	$T_{1/2}$
	(mg)	($\mu g/ml$)	(h)	($\mu g \cdot h/ml$)	(h)
Amoxicillin					
AMX/CA 500/125 mg	500	7.19 ± 2.26	1.5 (1.0-2.5)	53.5 ± 8.87	1.15 ± 0.20
Clavulanic acid					
AMX/CA 500 mg/125 mg	125	2.40 ± 0.83	1.5 (1.0-2.0)	15.72 ± 3.86	0.98 ± 0.12

AMX – amoxicillin, CA – clavulanic acid

* Median (range)

Amoxicillin and clavulanic acid serum concentrations achieved with amoxicillin/clavulanic acid are similar to those produced by the oral administration of equivalent doses of amoxicillin or clavulanic acid alone.

Distribution

About 25% of total plasma clavulanic acid and 18% of total plasma amoxicillin is bound to protein. The apparent volume of distribution is around 0.3-0.4 l/kg for amoxicillin and around 0.2 l/kg for clavulanic acid.

Following intravenous administration, both amoxicillin and clavulanic acid have been found in gall bladder, abdominal tissue, skin, fat, muscle tissues, synovial and peritoneal fluids, bile and pus. Amoxicillin does not adequately distribute into the cerebrospinal fluid.

No evidence for significant tissue retention of drug-derived material for either component has been reported in animals. Amoxicillin, like most penicillins, can be detected in breast milk. Trace quantities of clavulanic acid can also be detected in breast milk (see **USE IN SPECIAL POPULATIONS**).

Both amoxicillin and clavulanic acid have been reported to cross the placental barrier (see **USE IN SPECIAL POPULATIONS**).

Biotransformation

Amoxicillin is partly excreted in the urine as the inactive penicilloic acid in quantities equivalent to up to 10 to 25% of the initial dose. Clavulanic acid is extensively metabolized in man and eliminated in urine and faeces and as carbon dioxide in expired air.

Elimination

The major route of elimination for amoxicillin is via the kidney, whereas for clavulanic acid it is by both renal and non-renal mechanisms.

Amoxicillin/clavulanic acid has a mean elimination half-life of approximately one hour and a mean total clearance of approximately 25 l/h. Approximately 60 to 70% of the amoxicillin and approximately 40 to 65% of the clavulanic acid are excreted unchanged in urine during the first 6 h after administration of single amoxicillin/clavulanic acid 250 mg/125 mg or 500 mg/125 mg tablets. Various reported studies have found the urinary excretion to be 50-85% for amoxicillin and between 27-60% for clavulanic acid over a 24 hour period. In the case of clavulanic acid, the largest amount of drug is excreted during the first 2 hours after administration.

Concomitant use of probenecid delays amoxicillin excretion but does not delay renal excretion of clavulanic acid (see **DRUG INTERACTIONS**).

Age

The elimination half-life of amoxicillin is similar for children aged around 3 months to 2 years and older children and adults. For very young children (including preterm newborns) in the first week of life the interval of administration should not exceed twice daily administration due to immaturity of the renal pathway of elimination. Because elderly patients are more likely to have decreased renal function, care should be taken in dose selection, and it may be useful to monitor renal function.

Gender

Gender has no significant impact on the pharmacokinetics of either amoxicillin or clavulanic acid.

Renal impairment

The total serum clearance of amoxicillin/clavulanic acid decreases proportionately with decreasing renal function. The reduction in drug clearance is more pronounced for amoxicillin than for clavulanic acid, as a higher proportion of amoxicillin is excreted via the renal route. Doses in renal impairment must therefore prevent undue accumulation of amoxicillin while maintaining adequate levels of clavulanic acid (see **DOSE AND METHOD OF ADMINISTRATION**).

Hepatic impairment

Hepatically impaired patients should be dosed with caution and hepatic function monitored at regular intervals.

INDICATIONS

ENHANCIN SUSPENSION should be used in accordance with local official antibiotic-prescribing guidelines and local susceptibility data.

ENHANCIN SUSPENSION is indicated for the treatment of the following infections in adults and children (see **DOSE AND METHOD OF ADMINISTRATION** and **WARNINGS AND PRECAUTIONS**):

- Acute bacterial sinusitis (adequately diagnosed)
- Acute otitis media
- Acute exacerbations of chronic bronchitis (adequately diagnosed)
- Community acquired pneumonia
- Cystitis
- Pyelonephritis
- Skin and soft tissue infections in particular cellulitis, animal bites, severe dental abscess with spreading cellulitis.
- Bone and joint infections, in particular osteomyelitis.

Consideration should be given to official guidance on the appropriate use of antibacterial agents.

DOSE AND METHOD OF ADMINISTRATION

Doses are expressed throughout in terms of amoxicillin/clavulanic acid content except when doses are stated in terms of an individual component.

The dose of amoxicillin/clavulanic acid that is selected to treat an individual infection should take into account:

- The expected pathogens and their likely susceptibility to antibacterial agents (see **WARNINGS AND PRECAUTIONS**)
- The severity and the site of the infection
- The age, weight and renal function of the patient as shown below.

The use of alternative presentations of amoxicillin/clavulanic acid (e.g. those that provide higher doses of amoxicillin and/or different ratios of amoxicillin to clavulanic acid) should be considered as necessary (see **WARNINGS AND PRECAUTIONS**).

For children < 40 kg, this formulation of amoxicillin/clavulanic acid provides a maximum daily dose of 2400 mg amoxicillin/600 mg clavulanic acid, when administered as recommended below. If it is considered that a higher daily dose of amoxicillin is required, it is recommended that another preparation of amoxicillin/clavulanic acid is selected in order to avoid administration of unnecessarily high daily doses of clavulanic acid (see **WARNINGS AND PRECAUTIONS**).

The duration of therapy should be determined by the response of the patient. Some infections (e.g. osteomyelitis) require longer periods of treatment. Treatment should not be extended beyond 14 days without review (see **WARNINGS AND PRECAUTIONS** regarding prolonged therapy).

Adults and children ≥ 40 kg should be treated with the adult formulations of amoxicillin/clavulanic acid.

Children < 40 kg

Lower dose: 20 mg/5 mg/kg/day to 40 mg/10 mg/kg/day given in three divided doses.

Higher dose: 40 mg/10 mg/kg/day to: 60 mg/15 mg/kg/day given in three divided doses.

Body weight (kg)	20 mg/5 mg/kg/day Dose in ml to be given every 8 hours	40 mg/10 mg/kg/day Dose in ml to be given every 8 hours	60 mg/15 mg/kg/day Dose in ml to be given every 8 hours	Body weight (kg)	20 mg/5 mg/kg/day Dose in ml to be given every 8 hours	40 mg/10 mg/kg/day Dose in ml to be given every 8 hours	60 mg/15 mg/kg/day Dose in ml to be given every 8 hours
2.0	0.4	0.6	NR	21.0	2.8	5.6	8.4
3.0	0.4	0.8	NR	22.0	3.0	6.0	8.8
4.0	0.6	1.2	NR	23.0	3.2	6.2	9.2
5.0	0.8	1.4	NR	24.0	3.2	6.4	9.6
6.0	0.8	1.6	NR	25.0	3.4	6.8	10
7.0	1.0	2.0	NR	26.0	3.6	7.0	10.4
8.0	1.2	2.2	NR	27.0	3.6	7.2	10.8
9.0	1.2	2.4	NR	28.0	3.8	7.6	11.2
10.0	1.4	2.8	NR	29.0	4.0	7.8	11.6
11.0	1.6	3.0	NR	30.0	4.0	8.0	12
12.0	1.6	3.2	4.8	31.0	4.2	8.4	12.4
13.0	1.8	3.6	5.2	32.0	4.4	8.6	12.8
14.0	2.0	3.8	5.6	33.0	4.4	8.8	13.2
15.0	2.0	4.0	6	34.0	4.6	9.2	13.6
16.0	2.2	4.4	6.4	35.0	4.8	9.4	14
17.0	2.4	4.6	6.8	36.0	4.8	9.6	14.4
18.0	2.4	4.8	7.2	37.0	5.0	10.0	14.8
19.0	2.6	5.2	7.6	38.0	5.2	10.2	15.2
20.0	2.8	5.4	8	39.0	5.2	10.4	15.6

NR - Not recommended. There are no reported data for amoxicillin/clavulanic acid 4:1 formulations regarding doses higher than 40 mg/10 mg/kg per day in children under 2 years.

Alternative oral formulations of amoxicillin/clavulanic acid should be considered to deliver practical dose recommendations.

Children may be treated with amoxicillin/clavulanic acid tablets, suspensions or paediatric sachets. Children aged 6 years and below should preferably be treated with amoxicillin/clavulanic acid suspension or paediatric sachets.

The dose (ml) to be given to the patient three times daily can also be calculated using the following formula below:

$$\text{Dose (ml) given three times daily} = \frac{\text{Recommended amoxicillin* dose (mg/kg/day)} \times \text{weight (kg)}}{\text{Reconstituted amoxicillin * in suspension (mg/ml)} \times 3 \text{ (daily dose)}}$$

* Only consideration of the amoxicillin component is required for this calculation.

For example, a 6 kg child treated at 20 mg/5 mg/kg/day:

$$\begin{array}{l} \text{Dose (ml) given} \\ \text{three times daily} \end{array} = \frac{20 \text{ (mg/kg/day)} \times 6 \text{ (kg)}}{50 \text{ (mg/ml)} \times (\text{daily doses})}$$

$$\begin{array}{l} \text{Dose (ml) given} \\ \text{three times daily} \end{array} = \frac{120 \text{ (mg)}}{150 \text{ (mg/ml)}}$$

$$\begin{array}{l} \text{Dose (ml) given} \\ \text{three times daily} \end{array} = 0.8 \text{ ml}$$

Elderly

No dose adjustment is considered necessary. Elderly patients should be treated with adult formulations of amoxicillin/clavulanic acid.

Renal impairment

Dose adjustments are based on the maximum recommended level of amoxicillin

No adjustment in dose is required in patients with creatinine clearance (CrCl) greater than 30 ml/min.

Children < 40 kg

CrCl: 10-30 ml/min	15 mg/3.75 mg/kg twice daily (maximum 500 mg/125 mg twice daily).
CrCl < 10 ml /min	15 mg/3.75 mg/kg as a single daily dose (maximum 500 mg/125 mg).
Haemodialysis	15 mg/3.75 mg/kg per day once daily. Prior to haemodialysis 15 mg/3.75 mg/kg. In order to restore circulating drug levels, 15mg/3.75 mg per kg should be administered after haemodialysis.

Hepatic impairment

Dose with caution and monitor hepatic function at regular intervals (see **CONTRAINDICATIONS** and **WARNINGS AND PRECAUTIONS**).

Method of administration

Amoxicillin/clavulanic acid suspension is for oral use.

Amoxicillin/clavulanic acid suspension should be administered with a meal to minimise potential gastrointestinal intolerance.

Therapy can be started parenterally and continued with an oral preparation. Shake to loosen powder, add water as directed, invert and shake. Shake the bottle before each dose.

Instructions on reconstitution of the medicinal product before administration:

Bottles may be supplied with a ring-seal on the neck of the cap or with a removable foil-backed seal on the mouth of the bottle.

Check cap or bottle seal is intact before using. The cap ring-seal is broken once the cap is opened. Alternatively, if a foil-

backed seal on the mouth of the bottle is present, this should be removed at the time of preparation.

Shake bottle to loosen powder. Add water upto the mark on the bottle. Close, invert and shake well.
Shake the bottle well before each dose.

Any unused medicinal product or waste material should be disposed of in accordance with local requirements.

CONTRAINDICATIONS

- Hypersensitivity to the active substances, to any of the penicillins or to any of the excipients present in the formulation.
- History of a severe immediate hypersensitivity reaction (e.g. anaphylaxis) to another beta-lactam agent (e.g. a cephalosporin, carbapenem or monobactam).
- History of jaundice/hepatic impairment due to amoxicillin/clavulanic acid (see **UNDESIRABLE EFFECTS**).

WARNINGS AND PRECAUTIONS

Before initiating therapy with amoxicillin/clavulanic acid, careful enquiry should be made concerning previous hypersensitivity reactions to penicillins, cephalosporins or other beta-lactam agents (see **CONTRAINDICATIONS** and **UNDESIRABLE EFFECTS**).

Serious and occasionally fatal hypersensitivity reactions (including anaphylactoid and severe cutaneous adverse reactions) have been reported in patients receiving therapy with beta-lactams. Before initiating therapy with ENHANCIN, careful inquiry should be made concerning previous hypersensitivity reactions to penicillins, cephalosporins, carbapenems or other beta-lactam agents. If an allergic reaction occurs, **ENHANCIN** must be discontinued immediately and appropriate alternative therapy instituted.

These reactions are more likely to occur in individuals with a history of penicillin hypersensitivity and in atopic individuals.

In the case that an infection is proven to be due to an amoxicillin-susceptible organisms(s) then consideration should be given to switching from amoxicillin/clavulanic acid to amoxicillin in accordance with official guidance.

This presentation of amoxicillin/clavulanic acid is not suitable for use when there is a high risk that the presumptive pathogens have reduced susceptibility or resistance to beta-lactam agents that is not mediated by beta-lactamases susceptible to inhibition by clavulanic acid. This presentation should not be used to treat penicillin-resistant *S. pneumoniae*.

Convulsions may occur in patients with impaired renal function or in those receiving high doses (see **UNDESIRABLE EFFECTS**).

Amoxicillin/clavulanic acid should be avoided if infectious mononucleosis is suspected since the occurrence of a morbilliform rash has been associated with this condition following the use of amoxicillin.

Concomitant use of allopurinol during treatment with amoxicillin can increase the likelihood of allergic skin reactions.

Prolonged use may occasionally result in overgrowth of non-susceptible organisms.

The occurrence at the treatment initiation of a feverish generalised erythema associated with pustula may be a symptom of acute generalised exanthemous pustulosis (AGEP) (see **UNDESIRABLE EFFECTS**). This reaction requires amoxicillin/clavulanic acid discontinuation and contraindicates any subsequent administration of amoxicillin.

Amoxicillin/clavulanic acid should be used with caution in patients with evidence of hepatic impairment (see **DOSE AND METHOD OF ADMINISTRATION, CONTRAINDICATIONS** and **UNDESIRABLE EFFECTS**).

Hepatic events have been reported predominantly in males and elderly patients and may be associated with prolonged treatment. These events have been very rarely reported in children. In all populations, signs and symptoms usually occur during or shortly after treatment but in some cases may not become apparent until several weeks after treatment has ceased. These are usually reversible. Hepatic events may be severe and, in extremely rare circumstances, deaths have been reported. These have almost always been reported in patients with serious underlying disease or taking concomitant medications known to have the potential for hepatic effects (see **UNDESIRABLE EFFECTS**).

Antibiotic-associated colitis has been reported with nearly all antibacterial agents including amoxicillin and may range in severity from mild to life threatening (see **UNDESIRABLE EFFECTS**). Therefore, it is important to consider this diagnosis in patients who present with diarrhoea during or subsequent to the administration of any antibiotics. Should antibiotic-associated colitis occur, amoxicillin/clavulanic acid should immediately be discontinued, a physician be consulted and an appropriate therapy initiated. Anti-peristaltic medicinal products are contraindicated in this situation.

Periodic assessment of organ system functions, including renal, hepatic and haematopoietic function is advisable during prolonged therapy.

Prolongation of prothrombin time has been reported rarely in patients receiving amoxicillin/clavulanic acid. Appropriate monitoring should be undertaken when anticoagulants are prescribed concomitantly. Adjustments in the dose of oral anticoagulants may be necessary to maintain the desired level of anticoagulation (see **DRUG INTERACTIONS** and **UNDESIRABLE EFFECTS**).

In patients with renal impairment, the dose should be adjusted according to the degree of impairment (see **DOSE AND METHOD OF ADMINISTRATION**).

In patients with reduced urine output, crystalluria has been reported very rarely, predominantly with parenteral therapy. During the administration of high doses of amoxicillin, it is advisable to maintain adequate fluid intake and urinary output in order to reduce the possibility of amoxicillin crystalluria. In patients with bladder catheters, a regular check of patency should be maintained (see **OVERDOSE**).

During treatment with amoxicillin, enzymatic glucose oxidase methods should be used whenever testing for the presence of glucose in urine because false positive results may occur with non-enzymatic methods.

The presence of clavulanic acid in amoxicillin/clavulanic acid may cause a non-specific binding of IgG and albumin by red cell membranes leading to a false positive Coombs test.

There have been reports of positive test results using the Bio-Rad Laboratories Platelia *Aspergillus* EIA test in patients receiving amoxicillin/clavulanic acid who were subsequently found to be free of *Aspergillus* infection. Cross-reactions with non-*Aspergillus* polysaccharides and polyfuranoses with Bio-Rad Laboratories Platelia *Aspergillus* EIA test have been reported. Therefore, positive test results in patients receiving amoxicillin/clavulanic acid should be interpreted cautiously and confirmed by other diagnostic methods.

ENHANCIN powder for oral suspension contains 2.5 mg of aspartame (E951) per ml, a source of phenylalanine. This medicine should be used with caution in patients with phenylketonuria. Neither non-clinical nor clinical data are available to assess aspartame use in infants below 12 weeks of age.

This medicinal product contains maltodextrin (glucose). Patients with rare glucose-galactose malabsorption should not take this medicine.

Effects on ability to drive and use machines

Studies on the effects on the ability to drive and use machines have not been reported. However, undesirable effects may occur (e.g. allergic reactions, dizziness, convulsions), which may influence the ability to drive and use machines (see **UNDESIRABLE EFFECTS**).

DRUG INTERACTIONS

Oral anticoagulants

Oral anticoagulants and penicillin antibiotics have been widely used in practice without reports of interaction. However, in the reported literature there are cases of increased international normalised ratio in patients maintained on acenocoumarol or warfarin and prescribed a course of amoxicillin. If co-administration is necessary, the prothrombin time or international normalised ratio should be carefully monitored with the addition or withdrawal of amoxicillin. Moreover, adjustments in the dose of oral anticoagulants may be necessary (see **WARNINGS AND PRECAUTIONS** and **UNDESIRABLE EFFECTS**).

Methotrexate

Penicillins may reduce the excretion of methotrexate causing a potential increase in toxicity.

Probenecid

Concomitant use of probenecid is not recommended. Probenecid decreases the renal tubular secretion of amoxicillin. Concomitant use of probenecid may result in increased and prolonged blood levels of amoxicillin but not of clavulanic acid.

Mycophenolate mofetil

In patients receiving mycophenolate mofetil, reduction in pre-dose concentration of the active metabolite mycophenolic acid (MPA) of approximately 50% has been reported following commencement of oral amoxicillin plus clavulanic acid. The change in pre-dose level may not accurately represent changes in overall MPA exposure. Therefore, a change in the dose of mycophenolate mofetil should not normally be necessary in the absence of clinical evidence of graft dysfunction. However, close clinical monitoring should be performed during the combination and shortly after antibiotic treatment.

UNDESIRABLE EFFECTS

The most commonly reported adverse drug reactions (ADRs) are diarrhoea, nausea and vomiting.

The ADRs reported with amoxicillin/clavulanic acid, sorted by MedDRA System Organ Class are listed below.

The following terminologies have been used in order to classify the occurrence of undesirable effects.

Very common ($\geq 1/10$)

Common ($\geq 1/100$ to $< 1/10$)

Uncommon ($\geq 1/1,000$ to $< 1/100$)

Rare ($\geq 1/10,000$ to $< 1/1,000$)

Very rare ($< 1/10,000$)

Not known (cannot be estimated from the available data)

<u>Infections and infestations</u>	
Mucocutaneous candidosis	Common
Overgrowth of non-susceptible organisms	Not known
<u>Blood and lymphatic system disorders</u>	
Reversible leucopenia (including neutropenia)	Rare
Thrombocytopenia	Rare
Reversible agranulocytosis	Not known
Haemolytic anaemia	Not known
Prolongation of bleeding time and prothrombin time ¹	Not known
<u>Immune system disorders¹⁰</u>	
Angioneurotic oedema	Not known
Anaphylaxis	Not known
Serum sickness-like syndrome	Not known
Hypersensitivity vasculitis	Not known
<u>Nervous system disorders</u>	
Dizziness	Uncommon
Headache	Uncommon
Reversible hyperactivity	Not known
Convulsions ²	Not known
Aseptic meningitis	Not known
<u>Gastrointestinal disorders</u>	
Diarrhoea	Common
Nausea ³	Common
Vomiting	Common
Indigestion	Uncommon
Antibiotic-associated colitis ⁴	Not known
Black hairy tongue	Not known
Tooth discolouration ¹¹	Not known
<u>Hepatobiliary disorders</u>	
Rises in AST and/or ALT ⁵	Uncommon
Hepatitis ⁶	Not known
Cholestatic jaundice ⁶	Not known
<u>Skin and subcutaneous tissue disorders⁷</u>	
Skin rash	Uncommon
Pruritus	Uncommon
Urticaria	Uncommon
Erythema multiforme	Rare
Stevens-Johnson syndrome	Not known

Toxic epidermal necrolysis	Not known
Bullous exfoliative-dermatitis	Not known
Acute generalised exanthemous pustulosis (AGEP) ⁹	Not known
Drug reaction with eosinophilia and systemic symptoms (DRESS)	Not known
<u>Renal and urinary disorders</u>	
Interstitial nephritis	Not known
Crystalluria ⁸	Not known

¹ See **WARNINGS AND PRECAUTIONS**

² See **WARNINGS AND PRECAUTIONS**

³ Nausea is more often associated with higher oral doses. If gastrointestinal reactions are evident, they may be reduced by taking amoxicillin/clavulanic acid with a meal.

⁴ Including pseudomembranous colitis and haemorrhagic colitis (see **WARNINGS AND PRECAUTIONS**)

⁵ A moderate rise in AST and/or ALT has been reported in patients treated with beta-lactam class antibiotics, but the significance of these findings is unknown.

⁶ These events have been reported with other penicillins and cephalosporins (see **WARNINGS AND PRECAUTIONS**).

⁷ If any hypersensitivity dermatitis reaction occurs, treatment should be discontinued (see **WARNINGS AND PRECAUTIONS**).

⁸ See **OVERDOSE**

⁹ See **WARNINGS AND PRECAUTIONS**

¹⁰ See **CONTRAINDICATIONS** and **WARNINGS AND PRECAUTIONS**

¹¹ Superficial tooth discolouration has been reported very rarely in children. Good oral hygiene may help to prevent tooth discolouration as it can usually be removed by brushing.

USE IN SPECIAL POPULATIONS

Pregnancy

Limited reported data on the use of amoxicillin/clavulanic acid during pregnancy in humans do not indicate an increased risk of congenital malformations. In women with preterm, premature rupture of the foetal membrane it was reported that prophylactic treatment with amoxicillin/clavulanic acid may be associated with an increased risk of necrotising enterocolitis in neonates. Use should be avoided during pregnancy, unless considered essential by the physician.

Breastfeeding

Both substances are excreted into breast milk (nothing is known of the effects of clavulanic acid on the breast-fed infant). Consequently, diarrhoea and fungus infection of the mucous membranes are possible in the breast-fed infant, so that breast-feeding might have to be discontinued. The possibility of sensitisation should be taken into account. Amoxicillin/clavulanic acid should only be used during breast-feeding after benefit/risk assessment by the physician in charge.

OVERDOSE

Symptoms and signs of overdose

Gastrointestinal symptoms and disturbance of the fluid and electrolyte balances may be evident. Amoxicillin crystalluria, in some cases leading to renal failure, has been reported (see **WARNINGS AND PRECAUTIONS**).

Convulsions may occur in patients with impaired renal function or in those receiving high doses.

Amoxicillin has been reported to precipitate in bladder catheters, predominantly after intravenous administration of

large doses. A regular check of patency should be maintained (see **WARNINGS AND PRECAUTIONS**).

Treatment of intoxication

Gastrointestinal symptoms may be treated symptomatically, with attention to the water/electrolyte balance. Amoxicillin/clavulanic acid can be removed from the circulation by haemodialysis.

STORAGE

Store below 30°C, protected from moisture

The constituted suspension should be stored in a refrigerator at a temperature between 2 to 8°C and used within 7 days of constitution

SUPPLY

Bottle of 60 ml

Manufacturer:

Sun Pharmaceutical Ind. Ltd.
Industrial Area - 3,
Dewas - 455 001
INDIA

Product Registration Holder:

RANBAXY (MALAYSIA) SDN. BHD
(A Sun Pharma Company)
Lot 23, Bakar Arang Industrial Estate
08000 Sungai Petani.
Kedah, MALAYSIA

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